

1. Providers announce price schedules $p_A(t)$, $p_B(t)$

2. Users choose a provider and an execution period

Users and LLM applications



Coding · Dialogue · Writing · Data analysis



Flexible users

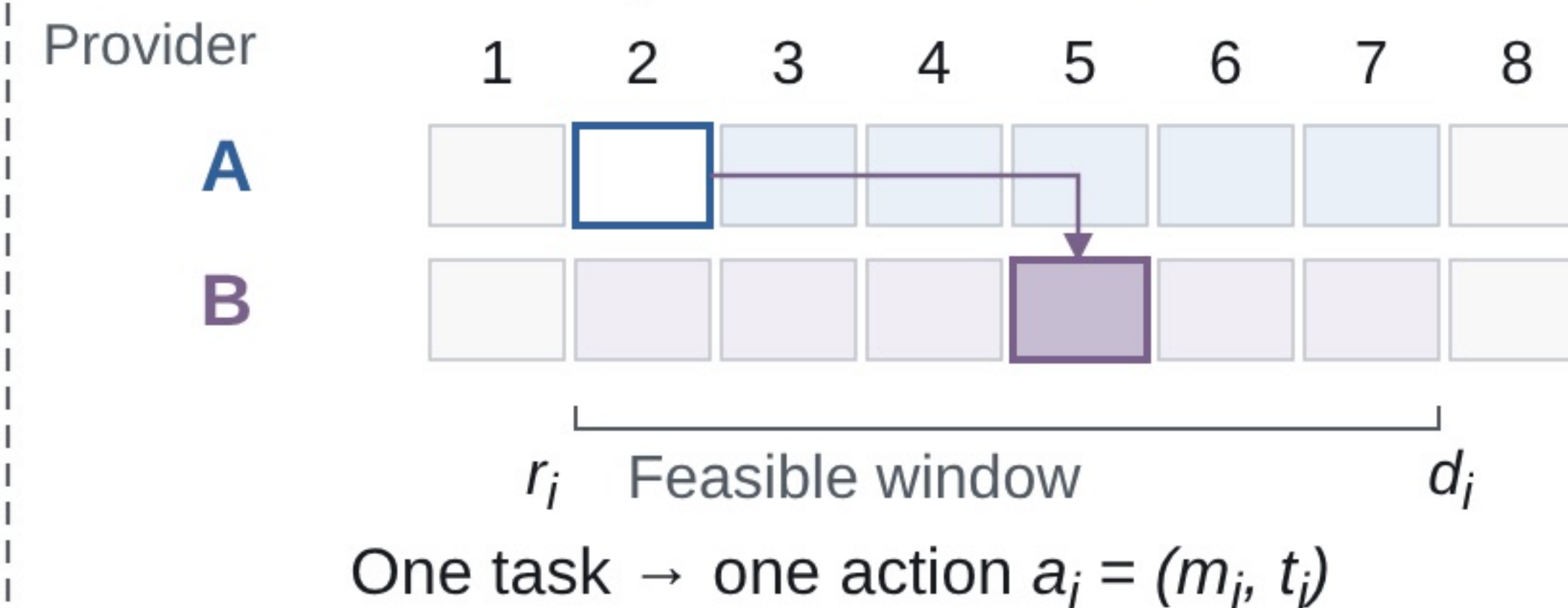
Choose within the feasible time window.



Time-rigid users

Fixed period; either provider is allowed.

Individual provider-time choice



Offline equilibrium computation



User parameters

r_i, d_i, b_i

Menus P_A, P_B and user costs

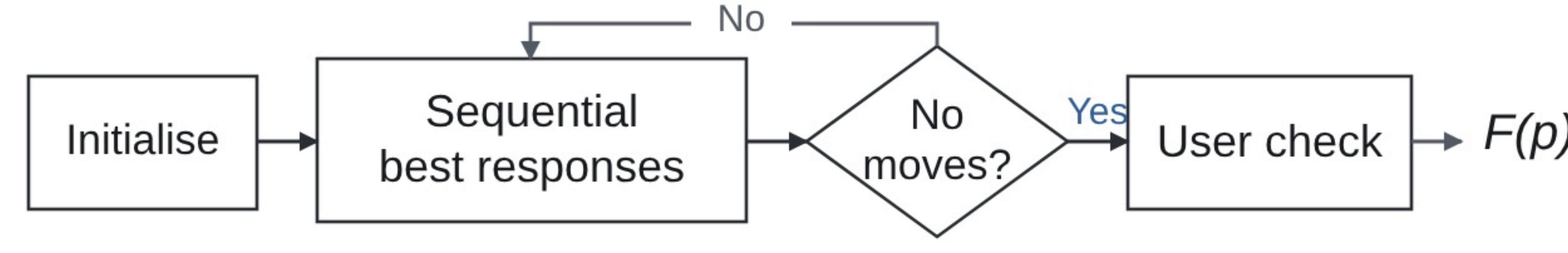
Bill · Delay · Switching · Congestion

1. Enumerate a price pair

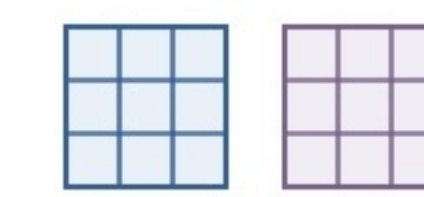
$(p_A, p_B) \in P_A \times P_B$

No — next price pair

2. Compute user equilibrium $F(p)$



3. Record provider payoffs



Populate A and B using $F(p)$

All pairs done?

Yes

4. Solve the provider game

Pure or mixed Nash strategies (x^*, y^*)

5. Verify unilateral deviations

Users: fixed prices · Providers: users re-equilibrate

Verified equilibrium (x^*, y^*, F)

Finite-menu stability; not a guarantee of universal welfare gains.

Competing LLM providers

Provider A



Price schedule

$p_A(t)$

K_A

$p_A(t) = \beta_A + \alpha_A z_t$

Price competition

Provider B



Price schedule

$p_B(t)$

K_B

$p_B(t) = \beta_B + \alpha_B z_t$

Precommitted time-of-use price schedules

K_m : reference capacity, not a hard limit.